FORMULA SHEET FOR NUMERICAL METHODS FOR ENGINEERS

Part 1: Modeling, Computers and Error Analysis

Error Definitions

True error: $E_t = true\ value - approximation$

True percent relative error:

$$\varepsilon_t = \frac{true\ value - approximation}{true\ value} \times 100\ \%$$

Approximate percent relative error:

$$\varepsilon_a = \frac{present\ approximation - previous\ approximation}{present\ approximation} \times 100\ \%$$

Stopping criterion:

Terminate computation when $\boldsymbol{\varepsilon_a} < \boldsymbol{\varepsilon_s}$ where $\boldsymbol{\varepsilon_s}$ is the desired percent relative error

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Taylor Series

Taylor series expansion

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f'''(x_i)}{3!}h^3 + \cdots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

where remainder

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!}h^{n+1} \quad \text{or} \quad R_n = O(h^{n+1})$$

Error propagation

For n independent variables $x_1, x_2, \ldots, x_n$ having errors $\Delta\tilde{x}_1, \Delta\tilde{x}_2, \ldots, \Delta\tilde{x}_n$, the error in the function f can be estimated via

$$\Delta f = \left|\frac{\partial f}{\partial x_1}\right|\Delta\tilde{x}_1 + \left|\frac{\partial f}{\partial x_2}\right|\Delta\tilde{x}_2 + \cdots + \left|\frac{\partial f}{\partial x_n}\right|\Delta\tilde{x}_n$$

Part 2: Roots of Equations

Method	**Formulation**	
Bisection	$x_r = \dfrac{x_l + x_u}{2}$	If $f(x_l) \times f(x_r) < 0$, set $x_u = x_r$ If $f(x_l) \times f(x_r) > 0$, set $x_l = x_r$
False Position	$x_r = x_u - \dfrac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$	If $f(x_l) \times f(x_r) < 0$, set $x_u = x_r$ If $f(x_l) \times f(x_r) > 0$, set $x_l = x_r$
Newton Raphson	$x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}$	
Secant	$x_{i+1} = x_i - \dfrac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$	

Part 3: Linear Algebraic Equations

Gauss Elimination

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & | & b_1 \\ a_{21} & a_{22} & a_{23} & | & b_2 \\ a_{31} & a_{32} & a_{33} & | & b_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} & | & b_1 \\ 0 & a'_{22} & a'_{23} & | & b'_2 \\ 0 & 0 & a''_{33} & | & b''_3 \end{bmatrix} \Rightarrow \begin{aligned} x_3 &= \frac{b''_{33}}{a''_{33}} \\ x_2 &= \frac{b'_2 - a'_{23}x_3}{a'_{22}} \\ x_1 &= \frac{b_1 - a_{12}x_2 - a_{13}x_3}{a_{11}} \end{aligned}$$

LU decomposition

Decomposition

Back Substitution

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix} \Rightarrow \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u'_{22} & u'_{23} \\ 0 & 0 & u''_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} \Rightarrow \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}$$

Forward Substitution

Gauss-Seidel method

$$\left.\begin{aligned} x_1^j &= \frac{b_1 - a_{12}x_2^{j-1} - a_{13}x_3^{j-1}}{a_{11}} \\ x_2^j &= \frac{b_2 - a_{21}x_1^{j} - a_{23}x_3^{j-1}}{a_{22}} \\ x_3^j &= \frac{b_3 - a_{31}x_1^{j} - a_{32}x_2^{j}}{a_{33}} \end{aligned}\right\} \begin{gathered} continue\ iteratively\ until \\ \left|\frac{x_i^j - x_i^{j-1}}{x_i^j}\right| 100\% < \varepsilon_s \\ for\ all\ x_i's \end{gathered}$$

With relaxation,

$$x_i^{new} = \lambda x_i^{new} + (1-\lambda)x_i^{old}$$

Part 4: Curve Fitting

Method	Formulation	Errors
Linear Regression	$y = a_0 + a_1 x$ where $S_r = \sum_{i=1}^{n}(y_i - a_0 - a_1 x_i)^2$ $a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$ $a_0 = \bar{y} - a_1\bar{x}$	$S_{y/x} = \sqrt{\frac{S_r}{n-2}}$ $r^2 = \frac{S_t - S_r}{S_t}$ where: $S_t = \sum (y_i - \bar{y})^2$
Polynomial Regression	For a 2nd order polynomial fit, $y = a_0 + a_1 x + a_2 x^2$ where $S_r = \sum_{i=1}^{n}(y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$ by differentiating $\boldsymbol{S_r}$ with respect to each coefficients and setting the partial derivatives equal to zero, we have: $\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{Bmatrix}$	$S_{y/x} = \sqrt{\frac{S_r}{n-(m+1)}}$ $r^2 = \frac{S_t - S_r}{S_t}$ where: $S_t = \sum (y_i - \bar{y})^2$

Multiple Linear Regression

$$y = a_0 + a_1 x_1 + \cdots + a_m x_m$$

$$S_{y/x} = \sqrt{\frac{S_r}{n-(m+1)}}$$

For a two-variable linear fit,

$$y = a_0 + a_1 x_1 + a_2 x_2$$

where

$$S_r = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_1 - a_2 x_2)^2$$

by differentiating $\boldsymbol{S_r}$ with respect to each coefficients and setting the partial derivatives equal to zero, we have:

$$\begin{bmatrix} n & \sum x_{1i} & \sum x_{2i} \\ \sum x_{1i} & \sum x_{1i}^2 & \sum x_{1i}x_{2i} \\ \sum x_{2i} & \sum x_{1i}x_{2i} & \sum x_{2i}^2 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} \sum y_i \\ \sum x_{1i}y_i \\ \sum x_{2i}y_i \end{Bmatrix}$$

$$r^2 = \frac{S_t - S_r}{S_t}$$

where:

$$S_t = \sum (y_i - \bar{y})^2$$

Newton's divided difference interpolating polynomial

For third order:

$$f_3(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$$

where

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

$$b_3 = f[x_3, x_2, x_1, x_0]$$

Lagrange interpolating polynomial

$$f_n(x) = \sum_{i=0}^{n} L_i(x) f(x_i)$$

where,

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

Part 6: Numerical Differentiation and Integration

A. Numerical Differentiation

Method	Formulation	Errors
Forward finite-divided difference	<u>First Derivative:</u> $f'(x_i) = \dfrac{f(x_{i+1}) - f(x_i)}{h}$ $f'(x_i) = \dfrac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$	$O(h)$ $O(h^2)$
Backward finite-divided difference	<u>First Derivative:</u> $f'(x_i) = \dfrac{f(x_i) - f(x_{i-1})}{h}$ $f'(x_i) = \dfrac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{2h}$	$O(h)$ $O(h^2)$
Centred finite-divided difference	<u>First Derivative</u> $f'(x_i) = \dfrac{f(x_{i+1}) - f(x_{i-1})}{2h}$ $f'(x_i) = \dfrac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2})}{12h}$	$O(h^2)$ $O(h^4)$

B. Numerical Integration

Method	Formulation
Trapezoidal rule	$I \cong (b-a)\dfrac{f(a) + f(b)}{2}$
Multiple-application trapezoidal rule	$I \cong \dfrac{(b-a)}{2n}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right)$

Simpson's 1/3 rule	$I \cong (b-a)\dfrac{f(x_0)+4f(x_1)+f(x_2)}{6}$
Multiple-application Simpson's 1/3 rule	$I \cong \dfrac{(b-a)}{3n}\left(f(x_0)+4\sum_{i=1,3,5,\ldots}^{n-1} f(x_i)+2\sum_{j=2,4,6,\ldots}^{n-2} f(x_j)+f(x_n)\right)$
Simpson's 3/8 rule	$I \cong (b-a)\dfrac{f(x_0)+3f(x_1)+3f(x_2)+f(x_3)}{8}$
Gauss Quadrature Gauss-Legendre	$I \cong c_0 f(x_0)+c_1 f(x_1)+\cdots+c_{n-1}f(x_{n-1})$ For two-point Gauss-Legendre: $c_0 = 1.0000000$ $c_1 = 1.0000000$ $x_0 = -0.577350269$ $x_1 = 0.577350269$ For three-point Gauss-Legendre: $c_0 = 0.5555556$ $c_1 = 0.8888889$ $c_2 = 0.5555556$ $x_0 = -0.774596669$ $x_1 = 0.0$ $x_2 = 0.774596669$ Change of variables: $x = \dfrac{(b+a)+(b-a)x_d}{2}$ $dx = \dfrac{b-a}{2}dx_d$

Part 7: Ordinary Differential Equations

Method	Formulation
Euler's First-Order RK	$y_{i+1} = y_i + k_1 h$ $k_1 = f(x_i, y_i)$
Heun's Second Order RK	$y_{i+1} = y_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h$ $k_1 = f(x_i, y_i)$ $k_2 = f(x_i + h, y_i + k_1 h)$
Midpoint Second Order RK	$y_{i+1} = y_i + k_2 h$ $k_1 = f(x_i, y_i)$ $k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h\right)$
Ralston's Second Order RK	$y_{i+1} = y_i + \left(\frac{1}{3}k_1 + \frac{2}{3}k_2\right)h$ $k_1 = f(x_i, y_i)$ $k_2 = f\left(x_i + \frac{3}{4}h, y_i + \frac{3}{4}k_1 h\right)$
Classical Fourth Order RK	$y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h$ $k_1 = f(x_i, y_i)$ $k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h\right)$ $k_3 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_2 h\right)$ $k_4 = f(x_i + h, y_i + k_3 h)$

Part 8: Partial Differential Equations

Method	Formulation
Elliptic PDEs Liebmann's Method	$T_{i,j} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}}{4}$
Parabolic PDEs (*one dimensional*) Explicit Method	$T_i^{l+1} = T_i^l + \lambda\left(T_{i+1}^l - 2T_i^l + T_{i-1}^l\right)$ $\lambda = k\,\frac{\Delta t}{(\Delta x)^2}$
Simple Implicit Method	$-\lambda T_{i-1}^{l+1} + (1+2\lambda)T_i^{l+1} - \lambda T_{i+1}^{l+1} = T_i^l$ $\lambda = k\,\frac{\Delta t}{(\Delta x)^2}$
Crank-Nicolson Method	$-\lambda T_{i-1}^{l+1} + 2(1+\lambda)T_i^{l+1} - \lambda T_{i+1}^{l+1} = \lambda T_{i-1}^l + 2(1-\lambda)T_i^l + \lambda T_{i+1}^l$ $\lambda = k\,\frac{\Delta t}{(\Delta x)^2}$